

IISER Mohali -PHY102-Electromagnetism

Assignment 3

Due date: 15th March 2025

Q1. Prove the Ohm's Law using fundamental concept of charge transport in conductor. Draw I vs V (DC) curve for (1) a simple resistor (R) circuit with V_a only. Can you mention a situation where the Ohm law is not valid and explain why? (2). Now consider a series resistor-capacitor (RC) circuit with applied constant voltage V_a . Derive relationship $Q = CV_a(1 - e^{-t/RC})$ for this RC circuit using boundary value problem. Assume you have switched in the circuit, $V_c=0$ at $t=0$, Assume $I = dQ/dt$, $V_a = 5$ volt, $R = 100$ Ohm, $C = 10\mu F$. Draw the curve for discharge of capacitor charges with time.

Q2. Calculate the current distribution through a cylindrical metal conductor of radius R when you have a non uniform current density ($j = ar^2$) across the cross section of conductor.

Q3. Suppose, there are 10^{24} free electron per cc in copper (Cu) wire. These free electrons contribute to a current of 1 ampere in the wire of 0.01cm^2 cross sectional area. Calculate the average drift speed of the electron in Cu and electric field. resistivity (ρ) of the copper is $= 1.7 \times 10^{-7}$

Q4. How will you prove that current conducting wire produces magnetic field? A charge particle of charge 'q' moving in space where both magnetic field 'B' and electric field 'E' are present. Write Lorentz force equation. What will happen if the charge particle enters perpendicular to both electric and magnetic field? In such situation, what are your conclusions under the conditions (a) $v=E/B$, (b) $v < E/B$, (c) $v > E/B$.

Q5. Find the magnetic field produced by an infinite long current (i) carrying wire at 'r' distance. How will you determine the direction of the magnetic field? what is ampere law and prove it using magnetic field produced by a infinite long current carrying wire.

Q6. what is Biot-Savart Law, write the expression for magnetic field for infinitesimal current carrying element 'dl' when it carries current 'i' and magnetic field measured at 'r' distance from the current carrying elemental length dl. Using above theory, estimate magnetic field at distance $r=5\text{cm}$ from an infinite long wire that carries current, $i= 1.2$ ampere.